# Driver Pulse - Product Design Document

## 1. Product Overview

Driver Pulse is a driver analytics system designed to provide transparent insights into driver performance, safety signals, and earnings progress. The system analyses multiple streams of operational data-trip logs, motion signals, audio intensity, and goal settings-to generate interpretable feedback that helps drivers understand how their driving behaviour and work patterns affect both safety and income outcomes.

**Execution Strategy**

**MVP Scope**

The MVP implements:

- Audio signal, motion detection
- earnings velocity forecasting
- ML goal prediction
- driver dashboard visualization

The product focuses on three core analytics modules:

1. **Earnings Velocity Module:** Evaluates if a driver is on pace to meet their financial targets.
2. **Audio Analysis Module:** Monitors the cabin environment for conflicts or excessive noise.
3. **Flagging System Module:** Detects and logs risky driving behaviours (braking, turning, acceleration).
4. **Engine Module:** Fuses motion and audio signals to quantify driver strain.

## 2. System Components Overview

The Driver Pulse system is organized into several logical layers:

**2.1 Data Ingestion**

Loads raw operational datasets including:

- trip records
- accelerometer signals
- audio intensity measurements

- driver goal configurations
- earnings velocity logs

This stage ensures consistent schemas and timestamp alignment before analytics processing begins.

**2.2 Signal Processing Layer**

Processes raw sensor data to detect driving events, audio_spikes such as:

**Motion Filtering:** Applies Butterworth low-pass filters to isolate vehicle maneuvers from engine vibrations.

- harsh braking
- rapid acceleration
- abnormal motion spikes

**Audio Smoothing:** Uses rolling window averages to distinguish sustained noise from momentary spikes.

- Normal
- Loud Cabin Noise
- Very Loud Cabin Noise
- Passenger Arguments

Detected events are stored as structured flags linked to specific trips and timestamps.

**2.3 Analytics Engine**

The analytics engine aggregates trip data, detected events, and driver goal information to compute higher-level metrics such as:

- stress scores
- trip quality ratings
- earnings velocity indicators

These outputs are stored as structured summaries for downstream visualization.

**2.4 Visualization Layer**

An interactive dashboard presents driver insights including:

- earnings progress
- goal pace indicators

- trip quality metrics
- flagged safety events
- overall trip history

The dashboard prioritizes clarity and interpretability, enabling drivers to quickly understand their operational status.

# 3. Earnings Velocity Module

### 3.1 Purpose

The Earnings Velocity module evaluates whether a driver is progressing toward a predefined earnings target within a specified working time window.

Drivers typically operate under uncertainty due to fluctuating demand, surge pricing, and traffic conditions. Without clear progress indicators, it is difficult for drivers to determine whether their current pace will allow them to achieve their earnings goals.

The Earnings Velocity system translates raw earnings data into a rate-based metric that enables continuous monitoring of goal progress.

### 3.2 Conceptual Model

The system treats earnings progress as a velocity comparison problem, analogous to motion dynamics.

Two primary rates are computed:

- Current earnings velocity
- Required earnings velocity

Comparing these values allows the system to determine whether the driver is ahead of schedule, on track, or behind target.

### 3.3 Velocity Delta

To determine goal progress, the system calculates the difference between the two velocity measures.

Formula:

current_velocity = cumulative_earnings / elapsed_hours

required_velocity = (target_earnings − cumulative_earnings) / remaining_hours

velocity_delta = current_velocity − required_velocity

Interpretation:

- Positive values indicate the driver is ahead of schedule.
- Values near zero indicate the driver is on pace.
- Negative values indicate the driver is behind target.

This delta forms the basis for the system's driver feedback logic.

### 3.4 Feature Inputs

The velocity system uses trip-level aggregates to compute supporting indicators, including:

- earnings_last_hour
- trips_last_hour
- average_fare_per_trip
- trip_duration
- distance_km
- surge_multiplier

Temporal context is captured using cyclic time features to reflect demand patterns throughout the day:

- hour_sin
- hour_cos

These features allow the system to contextualize earnings relative to operational conditions.

### 3.5 Driver Feedback Logic

Velocity delta is mapped into discrete feedback categories to simplify interpretation.

Typical categories include:

Ahead of target
Driver is earning faster than required.

On track
Driver is progressing at a rate sufficient to meet the goal.

Slightly behind
Driver is marginally below the required pace.

Significantly behind
Driver's current pace is insufficient to reach the goal without adjustment.

These feedback signals are surfaced through the driver dashboard.

**3.6 Edge Case Handling**

Several safeguards are included to maintain system stability.

- Zero Remaining Time

If remaining_hours equals zero, required_velocity becomes undefined. The system replaces zero values with NaN to prevent division errors.

- Early Shift Volatility

Very small elapsed_hours values can cause unstable velocity calculations. Early-stage smoothing prevents large fluctuations.

- Missing Goal Data

If a driver has not configured a goal, the velocity module is bypassed and standard performance metrics are displayed instead.

**3.7 Design Rationale**

- The system combines **deterministic velocity calculations** with **machine learning** to evaluate whether a driver is likely to meet their earnings goal.
- The deterministic component computes **current velocity, required velocity, and velocity delta**, providing transparent real-time indicators of driver progress.
- An **XGBoost classifier** is used to predict the probability of goal attainment using contextual features such as earnings rate, trip frequency, surge multiplier, and time-of-day patterns.
- XGBoost was selected because it **performs strongly on structured tabular data, captures nonlinear relationships between features, and provides fast inference suitable for near real-time systems**.
- To ensure robustness in environments with limited dependencies or connectivity, a **Logistic Regression model is implemented as an offline fallback**, providing lightweight predictive capability with minimal computational overhead.

# 4. Audio Analysis Module

### 4.1 Purpose

The **Audio Intensity Analysis Module** serves as the "ears" of the Driver Pulse system. While the Earnings Velocity module monitors financial health and the Flagging System monitors vehicle physics, this module evaluates the **human environment** inside the cabin. Its primary goal is to detect non-driving stress factors—such as passenger conflicts, loud music, or heavy traffic noise—that directly impact driver focus, safety, and passenger ratings.

### 4.2 Conceptual Model

Similar to how the Earnings module treats income as "velocity," this module treats cabin sound as a **weighted stressor**. It distinguishes between "Normal Operational Noise" (engine, wind) and "Acute Stressors" (shouting, loud disruptions).

The system doesn't just look at how loud it is; it looks at **what kind of noise** is occurring by fusing decibel levels with categorical classification.

### 4.3 The Scoring Model

The system calculates a real-time **Audio Event Score** to quantify the severity of the cabin environment.

**The Formula:**

Score = (0.6 * I{argument}) + (0.3 * I{loud}) + (0.1 * dB/100)

**Where:**

- **I_{argument} (Weight: 0.6):** A binary indicator (1 or 0) triggered by keyword detection or speech pattern analysis (e.g., "stop," "get out," "shouting"). Because conflict is the highest stressor, it carries the most weight.
- **$I_{loud} (Weight: 0.3):** A binary indicator triggered when the raw decibel level stays above **70 dB** for more than 3 seconds.
- **dB/100 (Weight: 0.1):** A continuous intensity factor that ensures that as volume increases, the score creeps higher even without an active argument.

### 4.4 Event Categories & Feedback Logic

The score is mapped into three actionable categories to provide clear feedback on the driver's dashboard:

| Category | Trigger Condition | Dashboard Feedback |
|---|---|---|
| **Passenger Argument** | Score $\> 0.35 + Speech Classification | " Alert: Potential Conflict Detected. Focus on safety." |
| **Very Loud Cabin** | Sustained Levels > 80 dB | " Warning: Excessive noise levels may cause fatigue." |
| **High Cabin Noise** | Sustained Levels > 70 dB | " Note: High ambient noise detected." |

### 4.5 Design Rational: Why Weighted Scaling?

- **False Positive Protection:** A single loud car horn outside the vehicle might hit 90 dB for a split second. By using a **3-second rolling average** and prioritizing the Argument_Flag, the system avoids penalizing drivers for outside traffic noise.
- **Explainability:** If a driver receives a 1-star rating, this module provides the evidence. The driver can point to a "Passenger Argument" flag at 10:15 PM to prove the low rating was due to a difficult passenger, not poor driving.
- **Sensor Fusion (The "Conflict" Trigger):** This module is most powerful when linked to the Accelerometer. If a **Sudden Brake** happens at the exact same time as a **Passenger Argument**, the system flags it as a "High-Stress Incident," which carries more weight in the final Trip Quality Rating than a simple braking event in a quiet car.

### 4.6 Edge Case Handling

- Privacy Preservation: To comply with privacy standards, the system does not record or store raw audio. It only processes the decibel levels and metadata (the "classification") in real-time, discarding the audio stream immediately after the score is calculated.

- Silence during Navigation: If the cabin is extremely quiet, the score stays near 0.1 (baseline ambient noise). This is interpreted as a "Low Stress" environment, contributing positively to the driver's overall health score.

# 5. Flagging System Module

### 5.1 Purpose

The Flagging System serves as the "physical observer" of the vehicle. By analyzing high-frequency accelerometer data, it provides objective evidence of driving style and road infrastructure health. This removes subjectivity from safety evaluations and identifies external stressors like poor road maintenance.

### 5.2 Detection Logic and Dynamics

The system monitors the three primary axes of motion. To ensure accuracy, raw signals are passed through a Butterworth Low-Pass Filter to remove high-frequency engine vibrations before applying detection thresholds.

**A. Longitudinal Events (a_x)**

Longitudinal acceleration tracks the vehicle's forward and backward force.

- **Sudden Acceleration:** Triggered when $a_x > 3.4$ m/s$^2$. This typically indicates aggressive driving or "jackrabbit" starts.
- **Sudden Braking:** Triggered when $a_x < -4.4$ m/s$^2$. This often correlates with distracted driving or late reactions to traffic.

**B. Lateral Events (a_y)**

Lateral acceleration tracks side-to-side force (centripetal force).

- **Sharp Turns:** Triggered when $|a_y| > 3.9$ m/s$^2$. This flags high-speed cornering which increases the risk of skidding and passenger discomfort.

**C. Vertical Events (a_z)**

The vertical axis is used to identify road anomalies.

- **Potholes:** Detected via sharp spikes in a_z after removing the constant 1g of gravity. To avoid false positives (such as closing a trunk or getting into the car), these are only logged when **Vehicle Speed > 5 km/h**.

### 5.3 Pothole Clustering

Individual a_z spikes are filtered through a **Spatial Deduplication** algorithm:

1. **Detection:** A vertical spike occurs at a specific GPS coordinate.

2. **Clustering:** The system groups events within a **0.0005-degree radius** (approx. 50 meters).
3. **Confidence Scoring:** A single report is flagged as a "Potential Anomaly." If multiple drivers report a spike at the same location, the "Pothole Confidence" increases, and the location is permanently marked on the fleet's road-condition map.

### 5.4 Design Rational: Why Physics-Based Threshold?

- Explainability: Drivers are not just told they are "unsafe"; they are shown the exact timestamp and force (e.g., "TRIP002 at 14:05: Sudden Braking at 5.2 $m/s^2$").
- Fairness: By using Adaptive Thresholds (comparing a_z spikes against current speed), the system avoids flagging a driver for hitting a bump at low speeds that would be dangerous at high speeds.
- Sensor Fusion: These flags are integrated with the Audio Module. A sharp turn occurring simultaneously with high cabin noise is escalated to a "High-Stress Conflict," whereas a sharp turn in a quiet cabin is treated as a routine maneuver.

# 6.Engine Module

### 6.1. Purpose

The Engine Module serves as the central intelligence hub of the Driver Pulse system. Its primary function is Sensor Fusion: the intelligent merging of mechanical telemetry (accelerometer) and environmental acoustics (audio) to calculate a unified Driver Stress Score.

By quantifying "strain," the system moves beyond simply counting mistakes and begins to understand the *context* of a driver's shift, distinguishing between a routine maneuver and a high-stress incident.

### 6.2. The Strain Fusion

The module operates on the principle that the risk level of a driving event is compounded by the environment in which it occurs. A sharp turn in a quiet cabin is a maneuver; a sharp turn during a shouting match is a safety risk.

The Stress Calculation Logic:

The system generates a Strain Index (S) for every trip by calculating the cross-correlation of events:

S = summation (Flag_motion * multiplier_audio)

- Baseline Strain: Derived from the frequency of sudden brakes and accelerations.
- Acoustic Multiplier: If the Audio Event Score is > 0.35 during a motion flag, the severity of that flag is multiplied by 1.5 times to 2.0 times.

# 6.Future Extensions

### 6.1. Predictive Earnings & Demand Intelligence

- **Demand Forecasting Models:** Utilizing historical trip densities and local event calendars to predict high-demand windows. This allows the "Required Velocity" calculation to adjust dynamically based on expected future volume rather than just current averages.
- **Surge Price Prediction:** Machine learning models trained on temporal and weather data to forecast surge multipliers. This helps drivers decide whether to stay online during a low-velocity period in anticipation of an imminent surge.
- **Geographic Demand Heatmaps:** Integrating GPS data to visualize "Earnings Hotspots."

### 6.2. Advanced Driver Support & Safety

- **Personalized Driver Recommendations:** The system will transition to a "Digital Coach" role. For example, if a driver's **Velocity Delta** is negative, the app might suggest: *"Demand is peaking 2km North; shifting there will increase your velocity by 15%."*
- **Stress-Based Fatigue Routing:** By fusing high **Stress Scores** (from the Audio and Flagging modules) with shift duration, the system can proactively recommend break times or suggest "low-stress" routes with fewer traffic lights and better road conditions to prevent burnout.
- **Real-time Pothole Avoidance:** Using the aggregated pothole database from the Flagging System to provide "Live Road Quality" alerts during navigation, reducing vehicle wear and tear.